Step 2: Fine-tuning. Variously called fine-tuning or alignment, this phase leverages proprietary methods or data to tweak model behaviors in ways model providers believe are helpful. For example, the LLama fine-tuning phase (Dubey et al., 2024) involves many rounds of reinforcement learning with human feedback (RLHF) to stamp out model negative behaviors, while Phi (Abdin et al., 2024) was fine-tuned on proprietary synthetic data to patch "gaps" in its mathematical reasoning abilities.

Reality: large-scale genAI models are (typically) fine-tuned using proprietary datasets/methods.

Step 3: Model updates. A key assumption of prior literature on model collapse is that models are *updated*. Updating generally involves training a new model using similar architecture initialized with weights from the prior step, techniques, and datasets to prior versions of the model. This matches reality—the literature documents numerous "families" of models that directly descend from one another (c.f. LLama 1, 2, and 3 (Touvron et al., 2023a;b; Dubey et al., 2024); GPT 1, 2, 3, 4 (Radford et al., 2018; Brown et al., 2020; Radford et al., 2019; Achiam et al., 2023); Phi 1.5, 2, 3 (Li et al., 2023; Abdin et al., 2023; 2024)). Prior works on model collapse have proposed several ways training data from prior model generations could be re-used (or not) during model updates.

The *replace* scenario, first proposed by Shumailov et al. (2024) assumes model trainers completely replace their training data at each update, using *only generated data output by the prior version of the model*. Schaeffer et al. (2025) point out that this edge case is interesting but impractical. The *accumulate* scenario (Gerstgrasser et al., 2024) assumes model trainers augment their original data with additional data that may contain AI-generated content at each update set. Finally, the *accumulate and subsample* scenario (Kazdan et al., 2025) echoes the accumulate scenario but subsamples a fixed-sized subset from the original and accumulated data at each update step (rather than letting dataset size grow linearly). This scenario acknowledges the real-world compute limitations.

We believe that the accumulate-and-subsample paradigm best reflects reality, so we will leverage it in our work. We support this opinion with evidence from three well-documented model families: Llama, GPT, and Phi. Table 2 records the training data used in publicly disclosed generations of these models. As the table shows, trainers re-use some prior training data for model updates, supplementing this with additional web content. Whitepapers for models published after 2023 do not contain specific training data information but suggest collection of additional online data at each update.

Reality: model trainers reuse data from prior generations and supplement this with new (internet-scraped) content when updating models.

However, most prior work on model collapse still overlooks a fundamental reality in model updates: *future models trained on internet-sourced content will be trained on outputs from other generative AI models*, not merely their own. Already, the internet is filled with generated content from various models (Sun et al., 2025). As model trainers collect new data to facilitate model updates, internet-sourced data will inevitably contain content from other generative models.

(Previously overlooked) reality: at each update step, models may be trained on *their own* and *other models'* generated outputs.

4 Formalizing An Iterative, Interactive Model Training Pipeline

Section 3 provides empirical evidence for two realities overlooked by prior studies: internet-scraped training datasets used for initial training may have substantial overlap, and models may be updated using *each others' generated outputs*. To study the effect of these two factors on model evolution, we propose a general workflow in which multiple entities regularly update their models using a mix of private, public, and generated data. Based on §3, we consider three types of training/update data:

- D_*: Public data used during initial training/updates by multiple entities (real data only).
- $\tilde{D}_k$: Private data used only by entity k for initial training/updates (real data only).
- $D_t = \{D_{t1}, D_{t2}, \dots D_{tk}\}$: Public data used for updates at time t by multiple entities (synthetic data). D_{tk} is data generated by the kth entity based on model $\theta_{t-1,k}$.

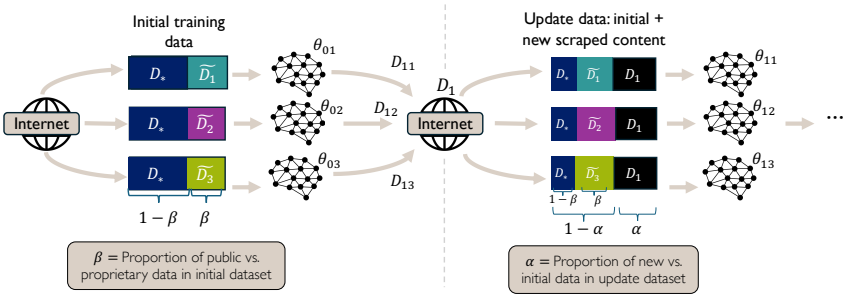

Figure 1: **Our dataset update scheme, parameterized by α and β.** *This paradigm best aligns with evidence from the literature given in §3 and strongly indicates that interactions between models, facilitating by training on others' generated data, are an important consideration for empirical and theoretical work on model evolution.*

Mapping these to realistic scenarios, D_* could be a public dataset like Common Crawl; $\tilde{D}_k$ could be a private dataset of math problems curated by entity k; and D_t could be an internet scrape from after initial model training. We weight the relative impact of these data types by the ratios α, β.

- β, $0 \leq \beta \leq 1$, is relative size of the initial public data set D_* compared to the initial private data set $\tilde{D}_k$. This fraction remains constant if/when initial data is reused for updates.
- α, $0 \leq \alpha \leq 1$ is the fraction of new data introduced at generation t, relative to the amount of initial data reused (following the "accumulate and subsample" paradigm of Kazdan et al. (2025)).

Interactive training workflow. We consider K entities, each seeking to train or update their own generative AI model. In the initial phase of training, denoted by time $t = 0$, each entity k trains its model based on a combination of a publicly available dataset D_* as well its own private dataset $\tilde{D}_k$. The trained model is represented generally by a parameter $\hat{\theta}_{t,k}$, i.e., $\hat{\theta}_{0,k} = \Phi_{0,k}(D_*, \tilde{D}_k)$, $k = 1, \ldots, K$. where each $\Phi_{0,k}$ represents a generic training algorithm. For model updates at stages $t > 0$, model parameters are updated via:

1. New public data D_t is generated uniformly at random using the most recent version of the models. Specifically, the data are sampled i.i.d. according to the mixture $\frac{1}{K}\sum_{k=1}^{K} P_{k,\hat{\theta}_{t-1,k}}$ where $P_{k,\theta}$ denotes the generative model used by k-th entity.
2. This data is placed online and collected by entities as training data for the next model update.
3. Each entity composes its training data for the next update, using a mix of the initial dataset $(D_*, \tilde{D}_k)$ and newly collected data D_t. Contributions from each dataset are weighted by α, β.
4. Each entity $k = 1, \ldots, K$ updates it model parameters via $\hat{\theta}_{t,k} = \Phi_{t,k}(\hat{\theta}_{t-1,k}, D_*, \tilde{D}_k, D_t)$. Here, $\Phi_{0,k}$ is a training algorithm that depends on the previous model parameter $\hat{\theta}_{k,t-1}$ as well as the data. As before, training may employ subsampling, weighting, and randomization.

In this workflow, entities interact through the release of publicly available synthetic data produced by prior generations of other entities' models. Thus, even though initial private training data are never shared, it could end up positively impacting other entities' models. This potential benefit of synthetic data sharing appears only in this interaction paradigm and has not be recognized in prior work.

5 Theory

We theoretically analyze the behavior of the interactive workflow. Similar to prior work (Gerstgrasser et al., 2024; Kazdan et al., 2025; Dey & Donoho, 2024; Dohmatob et al., 2025; Barzilai & Shamir, 2025), we focus on the linear regression models where each data point consists of a feature-response pair $(x, y) \in \mathbb{R}^d \times \mathbb{R}$. By the universality results of Dey & Donoho (2024), the analysis of this settings also applies to generalized linear models satisfying appropriate asymptotic normality assumptions.

Notation. For a $p \times q$ matrix A, we use A^+ to denote the Moore-Penrose pseudoinverse and $\mathsf{vec}(A)$ to denote the $pq \times 1$ vector obtained by stacking the columns. $\otimes$ denotes the Kronecker product. For $\alpha, \beta \in [0, 1]$ we set $\bar{\alpha} = 1 - \alpha$ and $\bar{\beta} = 1 - \beta$.

Training Workflow. We follow the training pipeline outlined in Section 4 in which K different models are trained on mixture of private, public, and generated data. At initialization, each entity

$k \in [K]$ combines its private data $\tilde{D}_k = (\tilde{x}_{ki}, \tilde{y}_{ki})_{i=1}^{\tilde{n}_k}$ with public data $D_* = (x_{*i}, y_{*i})_{i=1}^{n_*}$ to produce an estimate $\hat{\theta}_{k0}$ by minimizing the empirical loss

$$\textstyle\sum_{(x,y)\in\tilde{D}_k} \beta_0 L(x,y,\theta) + \sum_{(x,y)\in\tilde{D}_*} \bar{\beta}_0 L(x,y,\theta)$$

where $L(x,y,\theta) \coloneqq (y - x^\top\theta)^2$ is the squared error loss and $0 \le \beta_0 \le 1$ controls the relative weight placed on the private data. Training then proceeds for generation stages $t = 1, 2, 3, \ldots$ as follows:

1. Each entity k uses its most recent parameter estimate $\hat{\theta}_{t-1,t}$ to generate new data $D_{tk} = (x_{tki}, y_{tki})_{i=1}^{n_{tk}}$ according to the Gaussian model $y \mid x \sim \mathsf{N}(x^\top\theta_{t-1,k}, \sigma^2)$. The entire collection of generated samples is combined into a single public data set $D_t = \cup_{k=1}^K D_{tk}$.
2. Each entity k produces a new estimate $\hat{\theta}_{tk}$ by minimizing the empirical loss
$$\textstyle\sum_{(x,y)\in D_k} \bar{\alpha}_t\beta_t L(x,y,\theta) + \sum_{(x,y)\in D_*} \bar{\alpha}_t\bar{\beta}_t L(x,y,\theta) + \sum_{(x,y)\in D_t} \frac{\alpha_t}{K} L(x,y,\theta)$$
with weights $0 \le \alpha_t, \beta_t \le 1$.

We note that our framework could easily be extended to accommodate new, human-generated data at each time step, but we omit this in our formulation for analytic simplicity. Throughout our analysis we assume that all features are deterministic. We represent dataset $\tilde{D}_k$ with $\tilde{n}_k \times d$ matrix $\tilde{X}_k = [\tilde{x}_{k1}, \ldots, \tilde{x}_{k\tilde{n}_k}]^\top$ and $\tilde{n}_k \times 1$ vector $\tilde{y}_k = [\tilde{y}_{k1}, \ldots, y_{k\tilde{n}_k}]^\top$, and use the same convention for the public data (X_*, y_*) and the generated data (X_{tk}, y_{tk}). Data across different entities are then combined into "lifted" representations, which are denoted using boldface:

$$\tilde{\boldsymbol{X}} = \begin{bmatrix} \tilde{X}_1 & & \\ & \ddots & \\ & & \tilde{X}_K \end{bmatrix}, \quad \boldsymbol{y}_0 = \begin{bmatrix} \tilde{y}_1 \\ \vdots \\ \tilde{y}_K \end{bmatrix}, \quad \boldsymbol{X}_t = \begin{bmatrix} X_{t1} & & \\ & \ddots & \\ & & X_{tK} \end{bmatrix}, \quad \boldsymbol{y}_t = \begin{bmatrix} y_{tk} \\ \vdots \\ y_{tK} \end{bmatrix}$$

We note that information about which entity produced which sample is required for the analysis, but is not used during the training, where all data from the same generation are treated interchangeably.

Bias-Variance Decomposition. We derive exact formulas for the mean and variance of the estimators at each stage of the workflow. Given the features $(\tilde{\boldsymbol{X}}, X_*, \boldsymbol{X}_t)$ and learning weights (α_t, β_t) define

$$\tilde{\boldsymbol{S}} = \operatorname{diag}(\tilde{S}_1, \ldots, \tilde{S}_k) \coloneqq \tilde{\boldsymbol{X}}^\top\tilde{\boldsymbol{X}}, \qquad S_* \coloneqq X_*^\top X_* \qquad \boldsymbol{S}_t = \operatorname{diag}(S_{t1}, \ldots, S_{tk}) \coloneqq \boldsymbol{X}_t^\top\boldsymbol{X}_t$$

$$\boldsymbol{G}_t \coloneqq \bar{\alpha}_t\beta_t\tilde{\boldsymbol{S}} + \bar{\alpha}_t\bar{\beta}_t(\mathrm{I}_K \otimes S_*) + \alpha_t(\mathrm{I}_K \otimes \underline{\boldsymbol{S}}_t) \qquad \underline{\boldsymbol{S}}_t \coloneqq \tfrac{1}{K}\textstyle\sum_{k=1}^K S_{tk}$$

$$\boldsymbol{P}_t \coloneqq \bar{\alpha}_t\boldsymbol{G}_t^+\begin{bmatrix}\beta_t\tilde{\boldsymbol{S}} & \bar{\beta}_t(\mathbf{1}_K \otimes S_*)\end{bmatrix} \qquad \boldsymbol{Q}_t \coloneqq \alpha_t\boldsymbol{G}_t^+\Pi\boldsymbol{S}_t$$

where $\Pi \coloneqq \frac{1}{K}(\mathbf{1}_{K\times K} \otimes \mathrm{I}_d)$ is an orthogonal projection matrix and $\alpha_0 \equiv 0$.

Theorem 1. *Conditional on the initial data $D_0 \coloneqq (\tilde{D}_1, \ldots, \tilde{D}_K, D_*)$, the estimates $\hat{\boldsymbol{\theta}}_t = \mathsf{vec}(\hat{\theta}_{t1}, \ldots, \hat{\theta}_{tK})$ are Gaussian with mean and variance*

$$\mathbb{E}[\hat{\boldsymbol{\theta}}_t \mid D_0] = \boldsymbol{M}_t\begin{bmatrix}\tilde{\boldsymbol{X}}^+\tilde{\boldsymbol{y}} \\ X_*^+ y_*\end{bmatrix}, \qquad \mathsf{Cov}(\hat{\boldsymbol{\theta}}_t \mid D_0) = \boldsymbol{C}_t$$

where the matrices $\boldsymbol{M}_t$ and $\boldsymbol{C}_t$ are defined recursively with $\boldsymbol{M}_0 = \boldsymbol{P}_0$ and $\boldsymbol{C}_0 = \mathbf{0}_{Kd\times Kd}$ and

$$\boldsymbol{M}_t = \boldsymbol{P}_t + \boldsymbol{Q}_t\boldsymbol{M}_{t-1}, \qquad \boldsymbol{C}_t = \boldsymbol{Q}_t(\sigma^2\boldsymbol{S}_t^+ + \boldsymbol{C}_{t-1})\boldsymbol{Q}_t, \qquad t \ge 1.$$

Theorem 1 shows that the conditional mean of each estimate $\hat{\theta}_{tk}$ is a linear combination of the individual ordinary least squares (OLS) estimates $\tilde{X}_1^+\tilde{y}_1, \ldots, \tilde{X}_K^+\tilde{y}_K$ and $X_*^+ y_*$ for the private data and public data, respectively. For each generation t, similarity across entities can be assessed by comparing the rows of the $K \times (K+1)$ block partitioning of $\boldsymbol{M}_t$. At initialization, the off-diagonal blocks for the private data are zeroed out, but in later stages, these blocks become nonzero thereby allowing private data to be shared across entities. Homogenization (i.e., shrinkage towards a global consensus) occurs when row blocks are identical, and thus each entity has the same mean.

For our next result we mimic the experimental setup in Section 6 and assume that the initial data are generated from a Gaussian model with a common ground truth parameter and the heterogeneity across datasets arises from the differences in the features, i.e., the matrices $\tilde{S}_1, \ldots, \tilde{S}_K$.